

L1 cache

$$\frac{\text{num}}{\text{blocks}} = \frac{16 \cdot 1024}{16} = 2^{10}$$

$$\frac{\text{num}}{\text{sets}} = \frac{2^{10}}{4} = 2^8$$

$$\text{offset} = \log_2(16) = 4 \text{ bits}$$

$$\text{index} = \log_2(256) = 8 \text{ bits}$$

$$\text{tag} = 32 - (8 + 4) = 20 \text{ bits}$$

L2 cache

$$\text{block size} = 2^8$$

$$\frac{\text{num}}{\text{blocks}} = \frac{2^{20}}{2^8} = 2^{12}$$

$$\frac{\text{num}}{\text{sets}} = \frac{2^{10}}{4} = 2^8$$

$$\text{offset} = \log_2(256) = 8 \text{ bits}$$

$$\text{index} = \log_2(4096) = 12 \text{ bits}$$

$$\text{tag} = 32 - (12 + 8) = 12 \text{ bits}$$

1.

00000000010000001000 000001000000

L1

Tag = 00000000010000001000

Index =

- 00000000
- Block offset =
- 0100
- Miss coz empty
- So load into cache

- L2
Tag = 000000000100
Index = 000000001000
-
- Block offset = 01000000
- 0100
- Miss coz empty
- So load into cache

2.

00000000010000000110000001000000

L1
Tag = 00000000010000000110
Index = 00000000

-
- Block offset = 0100
-
- Miss new tag in this index
- So load into cache

- L2
Tag = 000000000100
Index = 000000000110
-
- Block offset = 01000000
-
- Miss coz new tag in this index
- So load into cache

3.

00000000010000001000000001000100

L1

tag= 000000000100000010000
index = 00000000

-
- Block offset = 0100
-
- Hit from 1
- So load into cache

L2

tag= 000000000100
index = 000000001000

-
- Block offset = 01000100
-

- Hit from 1
- So load into cache

2 hits 2 misses